Symmetries in Physics — Exercise Sheet 4

Schur's lemma and the group algebra

Variant: Questions only

Exercise 1 Use Schur's lemma to prove: (i) every irreducible complex representation of an abelian group is one-dimensional; (ii) in any irreducible representation, central elements act as scalars.

Exercise 2 Take the two-dimensional irreducible representation of $S_3 \cong D_3$ by rotations r^k through $2\pi k/3$ and reflections sr^k. Verify the orthogonality of matrix elements in two instances: $\sum_g |\Pi_{11}(g)|^2 = |G|/d$ and $\sum_g \Pi_{11}(g)\overline{\Pi_{12}(g)} = 0$.

Exercise 3 Granting the lecture's isomorphism $\mathbb{C}[G] \cong \bigoplus_i M_{d_i}(\mathbb{C})$, verify the dimension count $|G| = \sum_i d_i^2$ for S_3 and for D_4, and deduce how many times each irreducible occurs in the regular representation.

Exercise 4 On $\mathbb{Z}/n\mathbb{Z}$ define the convolution $(f * h)(x) = \sum_y f(y)h(x-y)$ and the Fourier transform $\widehat{f}(k) = \sum_x f(x)\omega^{-kx}$, $\omega = e^{2\pi \mathrm{i}/n}$. Prove the convolution theorem $\widehat{f * h} = \widehat{f}\,\widehat{h}$, and interpret it as the statement that the Fourier basis simultaneously block-diagonalises the regular representation.

Exercise 5 The quaternion group is $Q_8 = \{\pm 1, \pm \mathrm{i}, \pm \mathrm{j}, \pm \mathrm{k}\}$ with $\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = \mathrm{ijk} = -1$. (i) Find all one-dimensional representations. (ii) Deduce from $\sum d_i^2 = 8$ that exactly one two-dimensional irreducible remains, and realise it with Pauli matrices. (iii) Verify its irreducibility by a commutant argument.

Exercise 6 Verify, for D_4, the lecture's counting theorem: the number of irreducible representations equals the number of conjugacy classes.